

ZALAU, RomaniaWMO: **150630**Lat: **47.1950N**Lon: **23.0467E**Elev: **972**StdP: **14.19**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	17.7	4.4	7.1	16.7	9.2	9.1	21.4	19.5	33.0	16.7	33.2	2.4	140	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.5	89.2	68.6	86.4	67.9	83.4	66.9	70.9	84.0	69.6	82.6	68.3	80.3	4.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	102.8	75.4	65.5	97.6	73.9	64.1	93.0	73.2	35.3	84.2	34.1	82.8	33.1	80.6	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.7	13.6	11.1	7.9	92.3	4.5	4.1	4.6	95.3	2.0	97.7	-0.5	100.0	-3.8	103.0		
			7.0	74.0	4.1	1.6	4.1	75.1	1.6	76.1	-0.7	76.9	-3.7	78.1		

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	31.3	36.3	42.1	52.4	60.6	67.1	69.7	70.6	61.8	52.3	43.0	33.4	(6)
(7)		DBStd	15.60	7.73	8.06	7.89	8.35	6.50	6.05	5.37	6.19	8.02	7.71	8.44	7.71	(7)
(8)		HDD50	2135	581	385	262	72	5	0	0	0	5	68	242	515	(8)
(9)		HDD65	5380	1046	803	710	381	166	47	21	17	155	394	660	980	(9)
(10)		CDD50	2792	0	2	17	144	333	514	610	640	359	141	31	1	(10)
(11)		CDD65	560	0	0	0	3	29	111	166	191	58	2	0	0	(11)
(12)		CDH74	4908	0	0	0	32	223	903	1451	1834	450	15	0	0	(12)
(13)		CDH80	1534	0	0	0	2	21	234	445	711	121	0	0	0	(13)
(14)	Wind	WSAvg	3.6	3.4	3.7	4.2	4.5	3.9	3.5	3.3	3.3	3.5	3.2	3.8	3.4	(14)
(15)	Precipitation	PrecAvg	25.7	1.3	1.2	1.5	1.9	2.8	3.7	3.4	2.2	2.5	1.9	1.8	1.8	(15)
(16)		PrecMax	37.7	2.3	3.0	4.6	3.3	6.1	7.4	7.3	4.0	5.4	3.9	3.7	3.5	(16)
(17)		PrecMin	15.3	0.4	0.4	0.3	0.3	0.9	0.6	0.7	0.2	0.3	0.2	0.0	0.3	(17)
(18)		PrecStd	5.5	0.6	0.6	1.0	0.7	1.5	1.7	1.7	1.0	1.3	1.1	0.8	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	58.9	67.1	78.0	82.4	88.8	90.6	94.2	88.8	76.1	65.1	53.4	(19)
(20)			MCWB	45.6	48.1	52.2	58.9	63.1	70.8	69.0	68.9	68.1	60.2	54.8	46.6	(20)
(21)		2%	DB	47.1	54.0	61.5	73.3	79.2	85.8	87.9	91.0	82.9	72.9	61.7	48.9	(21)
(22)			MCWB	42.2	46.1	48.5	56.7	62.6	69.0	68.7	68.1	65.1	58.8	52.8	44.4	(22)
(23)		5%	DB	44.1	51.0	57.3	69.5	76.3	82.5	84.9	87.4	79.0	68.7	58.6	46.3	(23)
(24)			MCWB	40.3	44.4	46.6	54.7	61.1	67.1	67.8	67.5	63.4	57.5	51.3	42.4	(24)
(25)		10%	DB	41.1	48.2	53.7	65.9	73.0	79.3	81.9	83.7	75.0	64.2	55.3	43.2	(25)
(26)			MCWB	38.0	42.7	45.1	52.7	59.6	66.3	66.7	66.9	62.4	55.7	49.5	40.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	49.6	53.6	60.5	66.3	73.8	72.1	72.0	69.9	62.9	55.8	48.0	(27)
(28)			MCD B	50.4	57.5	65.5	75.1	78.3	85.0	84.8	87.2	86.0	71.8	63.2	51.8	(28)
(29)		2%	WB	42.9	47.0	49.9	58.2	64.1	70.6	70.6	70.3	66.7	60.4	53.6	44.9	(29)
(30)			MCD B	46.2	52.6	58.8	71.0	75.8	81.4	83.3	85.6	79.0	68.5	59.7	48.8	(30)
(31)		5%	WB	40.5	45.2	47.8	55.9	62.5	68.9	69.3	69.0	64.9	58.6	51.8	42.5	(31)
(32)			MCD B	43.6	50.1	55.6	67.0	73.4	79.9	81.4	83.4	76.1	67.1	57.8	45.6	(32)
(33)		10%	WB	38.3	43.3	45.8	54.1	60.9	67.3	68.1	67.7	63.1	56.7	50.0	40.8	(33)
(34)			MCD B	40.8	47.9	53.0	63.7	71.0	77.4	79.0	80.7	73.0	63.2	55.2	43.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.3	11.8	14.9	17.6	17.6	18.0	19.3	20.5	17.2	15.8	11.1	8.4	(35)
(36)			MCD BR	11.8	15.7	19.9	22.4	22.5	22.1	23.9	24.5	22.9	22.6	15.3	11.1	(36)
(37)			MCWBR	8.0	9.6	10.7	10.2	9.5	8.8	8.7	7.7	8.7	10.6	8.6	8.0	(37)
(38)		5% WB	MCD BR	11.0	14.5	17.4	20.9	20.2	20.1	21.2	22.2	20.5	19.2	14.1	10.6	(38)
(39)			MCWBR	7.7	9.3	10.2	10.1	9.4	8.8	8.3	7.9	8.4	9.8	8.0	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.319	0.346	0.386	0.434	0.420	0.418	0.437	0.438	0.401	0.371	0.344	0.319	(40)	
(41)		taud	2.324	2.278	2.221	2.120	2.225	2.261	2.219	2.213	2.282	2.343	2.364	2.358	(41)	
(42)		Ebn at Noon	238	256	262	259	268	269	262	256	254	244	226	224	(42)	
(43)		Edh at Noon	24	31	38	46	43	42	43	42	36	29	23	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	346	605	991	1406	1713	1875	1841	1691	1184	777	433	262	(44)	
(45)		RadStd	50	87	146	146	172	133	161	153	140	96	61	39	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)	
(47)	Regional (54 neighbors)	+1.24	N/A	N/A	+1.85	N/A	N/A	-241	-356	+202	+99	(47)

Nomenclature: See separate page